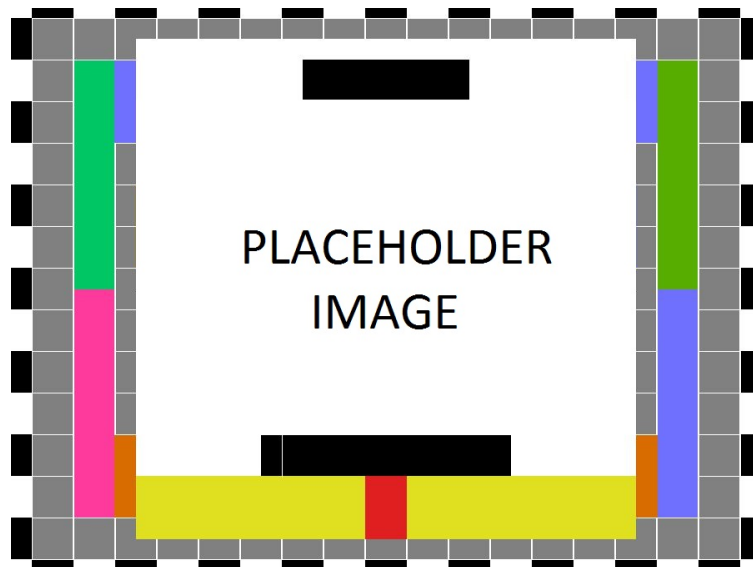


**DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
THE UNIVERSITY OF TEXAS AT ARLINGTON**

**ARCHITECTURAL DESIGN SPECIFICATION
CSE 4316: SENIOR DESIGN I
FALL 2026**



**TEAM NIMBUS
AQUAGUARD SMART WATER-LEAK MONITOR**

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REVISION HISTORY

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1.0	11.18.2026	AT, GH, CB	official release

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1 INTRODUCTION

Architecture of AquaGuard Smart Water-Leak Monitor as an ISO/IEC/IEEE 42010:2022 [1] description using IEEE 1016-2009 [2] viewpoints. Refines the SRS. Scope: the sensor device, the mobile app, and the cloud backend.

2 STAKEHOLDERS & ARCHITECTURAL CONCERNS

Stakeholder	Primary concerns
Homeowner (user)	fast reliable alerts, easy setup, battery life
Advisor	product viability, cross-platform parity, cost
Development team	feasibility, BLE reliability, testability
Support/maintainers	diagnosability, OTA update path

3 ARCHITECTURE VIEWPOINTS

Context (Section 4), Composition/Data Flow (Section 5), Logical/Functional (Sections 6–8), Deployment (Section 8). Style: an edge device (BLE peripheral) + mobile central + cloud, with detection at the edge and alerting split between local (BLE) and remote (push) paths. Interfaces #NN; subsystems [Layer] . [Name].

4 SYSTEM OVERVIEW & CONTEXT VIEW

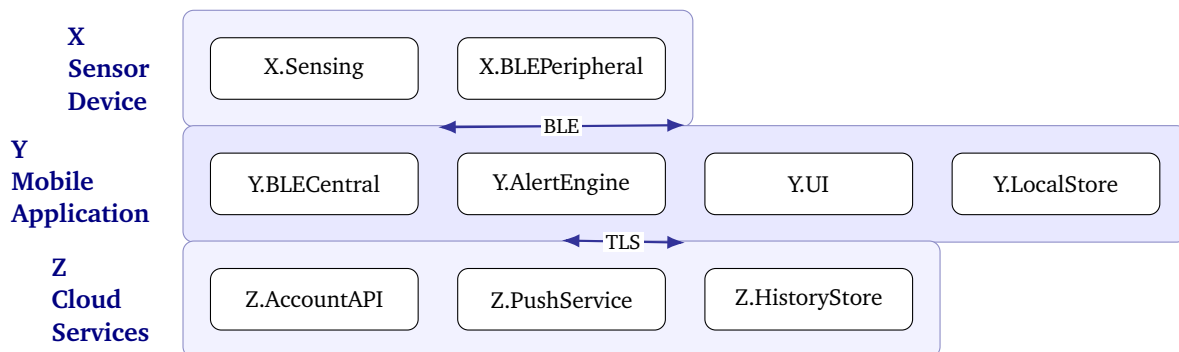


Figure 1: Three layers: sensor device, mobile app, cloud services

External actors: the homeowner (phone UI) and the platform push services (APNs/FCM). X senses and detects; Y connects, alerts locally, and relays; Z provides accounts, history, and remote push.

5 SUBSYSTEM DEFINITIONS & DATA FLOW

6 X LAYER SUBSYSTEMS — SENSOR DEVICE

6.1 X.SENSING

Responsibilities: sample moisture/flow and run on-device threshold detection (FR-01) at low power (PE-01).

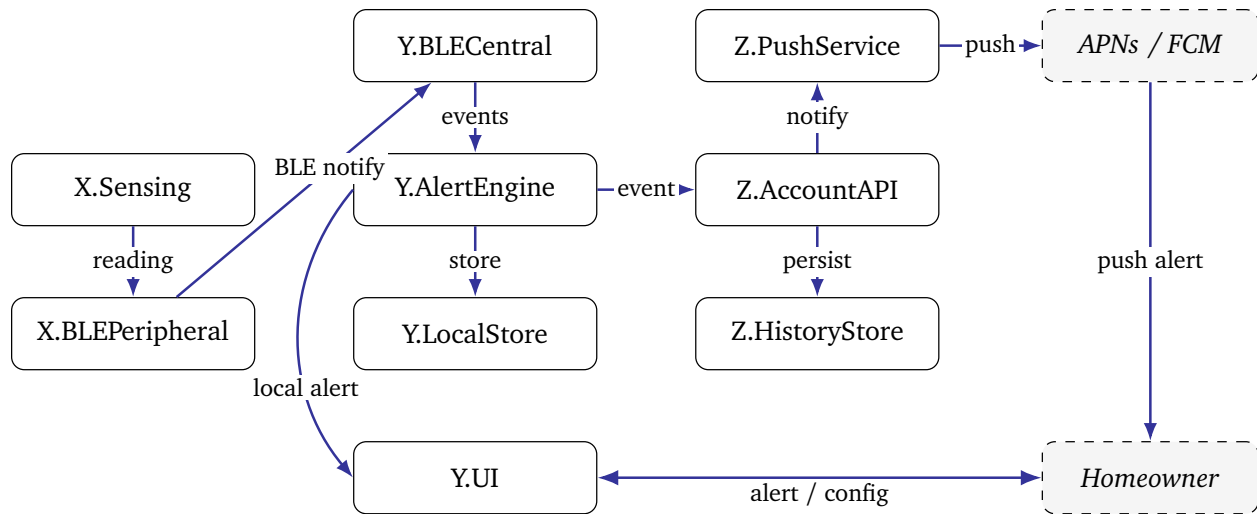


Figure 2: Sensor → BLE → app → (local alert) / cloud → push

ID	Description	Inputs	Outputs
#01	Sensor sampling	moisture/flow	reading, leak flag

6.2 X.BLEPERIPHERAL

Responsibilities: expose the GATT profile (EI-01) — live reading, leak notify, threshold config, battery — and manage bonded, encrypted links (SE-01). **Rationale:** on-device detection keeps alerting working even if the phone is briefly away and saves radio energy (AD-01).

7 Y LAYER SUBSYSTEMS — MOBILE APPLICATION

7.1 Y.BLECENTRAL

Responsibilities: scan, pair/bond, subscribe to notifications, read/write config (FR-02).

ID	Description	Inputs	Outputs
#02	BLE central link	GATT notifications	events, config writes

7.2 Y.ALERTENGINE

Responsibilities: raise local visual/audible/haptic alerts (FR-03) and forward events to the backend for remote push when disconnected (FR-04).

7.3 Y.UI & Y.LOCALSTORE

Y.UI: guided setup, live status, history (US-01, FR-05). **Y.LocalStore:** on-device event/config cache; source for offline history.

8 Z LAYER SUBSYSTEMS — CLOUD SERVICES

8.1 Z.ACCOUNTAPI

Responsibilities: device registration, user account, event ingestion over TLS (SE-01). **Deployment:** small managed backend + database.

8.2 Z.PUSHSERVICE

Responsibilities: deliver alerts via APNs/FCM (EI-02, FR-04).

8.3 Z.HISTORYSTORE

Responsibilities: durable event history (FR-05) synced with the app.

9 ARCHITECTURE DECISIONS & RATIONALE

9.1 AD-01: EDGE DETECTION ON THE DEVICE

Context: FR-01 latency + PE-01 battery + intermittent phone presence. **Decision:** detect on the device, not in the app/cloud. **Alternatives:** stream raw readings to the phone (higher radio energy, fails when away). **Consequences:** firmware complexity up, energy/latency improved. **Affects:** FR-01, PE-01, FR-04.

9.2 AD-02: DUAL LOCAL/REMOTE ALERT PATHS

Context: FR-03 (nearby) and FR-04 (away). **Decision:** local BLE alert plus cloud push fallback. **Alternatives:** push-only (slow when nearby), BLE-only (no reach when away). **Consequences:** two paths to test. **Affects:** FR-03, FR-04.

10 REQUIREMENTS TRACEABILITY

Req ID	Requirement (abbrev.)	Realizing subsystem(s)
FR-01	Leak detection	X.Sensing
FR-02	Pairing/config	Y.BLECentral, X.BLEPeripheral
FR-03	Local alert	Y.AlertEngine
FR-04	Remote alert	Y.AlertEngine, Z.PushService
FR-05	History	Y.LocalStore, Z.HistoryStore
SE-01	Secure pairing/TLS	X.BLEPeripheral, Z.AccountAPI

REFERENCES

- [1] ISO/IEC/IEEE, “ISO/IEC/IEEE 42010:2022 — Software, systems and enterprise — Architecture description,” International Organization for Standardization, Geneva, Switzerland, Tech. Rep., 2022.
- [2] IEEE, “IEEE Std 1016-2009 — IEEE Standard for Information Technology — Systems Design — Software Design Descriptions,” Institute of Electrical and Electronics Engineers, New York, NY, USA, Tech. Rep., 2009.